

Impact of Chromatic Dispersion on the System Limitation Due to Polarization Mode Dispersion

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Abstract—The impact of chromatic dispersion on the system limitation due to polarization mode dispersion (PMD) is investigated through Monte Carlo simulations. The waveplate model is used to realistically simulate the variation of PMD with frequency. It is shown that the effects of higher order PMD are compounded in the presence of chromatic dispersion. This can lead to a reduction in the PMD limit for large values of chromatic dispersion. However, it does not lead to a reduction in the PMD limit for the moderate levels of chromatic dispersion that would be encountered in most systems.

Index Terms—Optical fiber communications, optical fiber dispersion, polarization mode dispersion.

I. INTRODUCTION

POLARIZATION mode dispersion (PMD) is an important source of impairment in high-capacity optical communication systems. In the first-order approximation for PMD, a pulse at the fiber input can be decomposed into pulses aligned with the two principal states of polarization (PSP's) [1]. At the fiber output these pulses are relatively delayed in time by the differential group delay (DGD), resulting in a distorted output pulse. The DGD and the PSP's are both functions of frequency; but in the first-order approximation they are assumed constant over the signal spectrum. In the absence of chromatic dispersion, the PMD limitation given by the first-order approximation is essentially the same as the limit found when higher order PMD is also considered [2], [3].

It has been shown that the effects of higher order PMD are compounded in the presence of chromatic dispersion [4], [5]. It is, therefore, expected that the combined effects of PMD and chromatic dispersion when treated together may be more severe than when they are treated separately. This is an important consideration since many systems will have both impairments. However, we have not seen results that indicate whether the PMD limit is in fact significantly affected by the presence of chromatic dispersion. In this letter, we examine, through simulations, the distortion of unchirped Gaussian input pulses due to the combined effects of PMD and chromatic dispersion in order to determine the impact of chromatic dispersion on the PMD limitation. The waveplate model is used to simulate the PMD, and to aid in the analysis, first-order PMD is also investigated.

II. SIMULATION METHOD

The waveplate model [6] is used in the simulations to give a realistic frequency-dependent PMD. Each fiber consists of 500 polarization maintaining fiber sections with randomly oriented birefringence axes. The delays between fast and slow axes are uniformly distributed in $(0, \Delta\tau_{\max})$ where $\Delta\tau_{\max}$ determines the mean DGD of a statistical ensemble of fibers.

We consider the transmission of both 010 and 101 bit patterns which have electric fields at the fiber input given by $E_1(t)$ and $E_2(t)$, respectively, where $E_1(t)$ is taken as $E_1(t) = E_0 \exp[-1/4(t/\sigma_i)^2 - i\omega_0 t]e$ and $E_2(t)$ is taken as $E_2(t) = E_1(t + T) + E_1(t - T)$. E_0 is a normalization constant, ω_0 is the pulse carrier frequency, e is the Jones vector representing the input state of polarization, T is the pulse period, and σ_i is the rms width of an isolated optical pulse at the fiber input, given by $P_i(t) = |E_1(t)|^2$. The optical pulse forms at the fiber output for the 010 and 101 patterns are found from $P_{1,2}(t) = |A_{1,2}(t)|^2$, where $A_1(t)$ and $A_2(t)$ are the slowly varying amplitudes of the pulse envelope [7] for the 010 and 101 patterns, respectively, with

$$A_{1,2}(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} d(\Delta\omega) S_{1,2}(\Delta\omega) \cdot \exp[i(0.5\beta_2 L(\Delta\omega)^2 - \Delta\omega t)] M(\omega) e$$

where ω is the angular frequency; $\Delta\omega = \omega - \omega_0$; $S_1(\omega)$ is the the Fourier transform of $[P_i(t)]^{1/2}$; $S_2(\omega)$ is the the Fourier transform of $[P_i(t + T)]^{1/2} + [P_i(t - T)]^{1/2}$; M is the Jones matrix of the fiber; L is the fiber length; $\beta_2 = d^2\beta/d\omega^2$ accounts for chromatic dispersion, where β is the propagation constant.

To characterize the pulse distortion we define an eye opening B at decision time $\bar{t}$ by

$$B = P_1(\bar{t}) - P_2(\bar{t})$$

where $\bar{t}$ is the mean transit time of an isolated pulse, given by $\int_{-\infty}^{\infty} t P_1(t) dt$. In our simulations T is taken as 100 ps, which corresponds to a bit rate of 10 Gb/s, and σ_i is taken as 25 ps. The results can be generalized in terms of τ/T , where τ is the mean value of DGD, if the ratio $\sigma_i/T = 1/4$ is maintained and the values of $\beta_2 L$ are scaled appropriately. We define a power penalty due to PMD as

$$\delta = -10 \log [B(\beta_2 L)/B_0(\beta_2 L)]$$

where $B_0(\beta_2 L)$ is the eye opening with chromatic dispersion but without PMD and $B(\beta_2 L)$ is the eye opening with both

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chromatic dispersion and PMD. We also consider an effective rms pulse broadening due to PMD given by [8], [9]

$$\sigma_{\text{eff}}^2 = \sigma^2(\beta_2 L) - \sigma_0^2(\beta_2 L)$$

where $\sigma_0(\beta_2 L)$ is the rms output pulsewidth with chromatic dispersion but without PMD, and $\sigma(\beta_2 L)$ is the rms output pulsewidth with both chromatic dispersion and PMD. $P_1(t)$ is used in the calculation of $\sigma_0(\beta_2 L)$ and $\sigma(\beta_2 L)$. The power penalty and the effective pulse broadening due to chromatic dispersion here are those found when chromatic dispersion acts alone. Any further penalty or pulse broadening, part of which may be due to the interaction of chromatic dispersion and higher order PMD, is attributed to PMD. In this way, the PMD limitation may depend on the value of the chromatic dispersion.

The PMD limitation, for a given value of chromatic dispersion, is determined by the mean value of the DGD, τ . We account for the statistical nature of PMD by examining six statistical ensembles, each consisting of 5000 fibers, with values of τ ranging from $0.075T$ ($\Delta\tau_{\text{max}} = 0.63056$ ps) to $0.175T$ ($\Delta\tau_{\text{max}} = 1.47131$ ps). For each fiber, $M(\omega)$ is calculated at discrete frequencies centered around $\omega_0 = 1216$ rad/ps (corresponding to a wavelength of about 1550 nm) and $P_1(t)$ and $P_2(t)$ are calculated for eight values of $\beta_2 L$ ranging from 0 to -2000 ps². The power penalty δ is then calculated using $P_1(t)$ and $P_2(t)$ and the effective pulse broadening σ_{eff}^2 is calculated using $P_1(t)$. The input state of polarization $\mathbf{e}$ is $(\begin{smallmatrix} 0 \\ 1 \end{smallmatrix})$ throughout the simulations.

For each fiber and value of $\beta_2 L$, δ and σ_{eff}^2 are also calculated using the first-order approximation for PMD. For each fiber, $P_1(t)$ and $P_2(t)$ are calculated with all frequencies having the same PSP's and DGD as at the pulse carrier frequency ω_0 . Six additional ensembles of 5000 fibers were examined for first-order PMD, with values of τ up to $0.33T$.

III. RESULTS

To observe the effects of the interaction between chromatic dispersion and higher order PMD the statistics of σ_{eff}^2 have been examined for the ensemble of fibers with $\tau = 0.15T$. The behavior here is typical of the six ensembles of fibers we have examined and is in agreement with results in [4] and [5]. For first-order PMD, σ_{eff}^2 is determined by the PMD and does not depend on the chromatic dispersion. The mean and standard deviation of σ_{eff}^2 for first-order PMD are in close agreement with the values obtained for the waveplate model with no chromatic dispersion; the mean and standard deviation of σ_{eff}^2 are 43.5 and 43.3 ps², respectively, for first-order PMD and 44.2 and 42.9 ps² for the waveplate model with $\beta_2 L = 0$. Incidentally, the simulation results for first-order PMD are close to the expected values; it can be shown [10] that the mean and standard deviation of σ_{eff}^2 with first-order PMD are both equal to $\pi\tau^2/16$. The mean value of σ_{eff}^2 for the waveplate model is only weakly dependent on chromatic dispersion; the mean values of σ_{eff}^2 range from 44.0 to 44.2 ps² for the eight values of $\beta_2 L$. However, the standard deviation in σ_{eff}^2 increases with chromatic dispersion as shown in Fig. 1. So, in a statistical sense, the effective pulse broadening due to PMD worsens as chromatic dispersion increases due to the interaction of chromatic dispersion and higher order PMD.

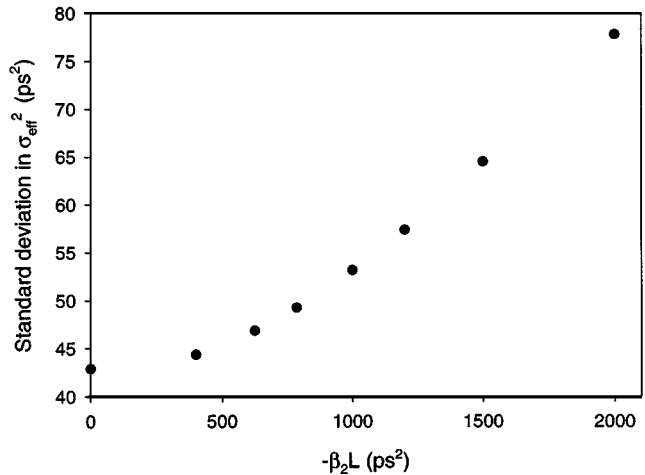


Fig. 1. Standard deviation in the effective pulse broadening as a function of chromatic dispersion, for the waveplate model with $\tau = 0.15T$.

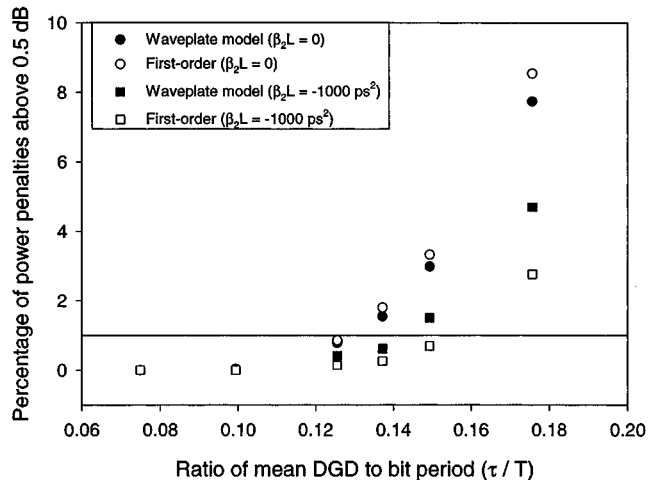


Fig. 2. Percentage of power penalties above 0.5 dB versus the ratio of mean DGD to bit period with $\beta_2 L = 0$ and $\beta_2 L = -1000$ ps².

Fig. 3 shows the limiting PMD values as a function of chromatic dispersion for both the waveplate model and the first-order approximation for PMD. To understand the results for the waveplate model it is useful to first examine the results for first-order PMD. The PMD limit for the first-order approximation increases with $|\beta_2 L|$. To explain this result we note that if the effective pulse broadening is the same for two different values of chromatic dispersion, the fractional broadening $\sigma(\beta_2 L)/\sigma_0(\beta_2 L)$ is lower for the larger value of chromatic dispersion. So there is a tendency for lower power penalties with higher values of chromatic dispersion. With first-order PMD it is useful to imagine input pulses of different widths [given by $\sigma_0(\beta_2 L)$] which propagate in a fiber with only first-order PMD. Due to the sensitivity of the penalty to the pulse shape, the broader pulses (with higher chromatic dispersion) tend to have a lower PMD penalty. This effect has been observed in experiments with first-order PMD [5].

We consider the system to be limited by PMD when the power penalty δ is above 0.5 dB with a probability greater than one percent. Results are presented in Fig. 2 for two values of chromatic dispersion: $\beta_2 L = 0$ and $\beta_2 L = -1000$ ps². With $\beta_2 L = 0$,

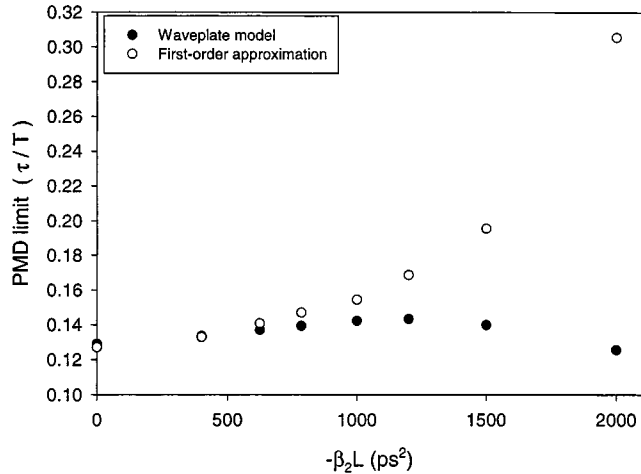


Fig. 3. PMD limitation as a function of chromatic dispersion for the waveplate model and the first-order approximation of PMD.

the limiting PMD value is about $0.129T$ for the waveplate model and $0.127T$ for the first-order approximation. So in the absence of chromatic dispersion the PMD limits for first-order PMD and the waveplate model are very close. We have estimated the error in the calculation of the PMD limit for the waveplate model with $\beta_2 L = 0$. Each 5000-fiber ensemble was divided into five 1000-fiber ensembles and five plots were obtained for the percentage of penalties above 0.5 dB versus τ/T . Five PMD limits were then obtained from these plots. The five PMD limits had a mean value of $0.129T$ and a standard deviation of $0.002T$. We, therefore, estimate the error in the determination of the PMD limits to be less than two percent.

With frequency-dependent PMD there are two conflicting effects. For equal values of effective pulse broadening there is a lower fractional broadening for higher values of chromatic dispersion and a tendency for lower penalties. However, the standard deviation in the effective pulse broadening increases with chromatic dispersion due to the interaction of chromatic dispersion and higher order PMD and this effect tends to increase the penalties for higher values of chromatic dispersion. Similar to first-order PMD, the PMD limit for the waveplate model initially increases with $|\beta_2 L|$ until a value of about 1200 ps^2 . Then the interaction of chromatic dispersion and higher order PMD begins to dominate and the PMD limit starts to decrease with increasing $|\beta_2 L|$.

With an input rms pulsewidth of 25 ps, a system would normally be designed with $|\beta_2 L| \leq 1000 \text{ ps}^2$ (the fractional pulse broadening and power penalty due to chromatic dispersion alone were 1.28 and 1.2 dB with $\beta_2 L = -1000 \text{ ps}^2$). The interaction between chromatic dispersion and higher order PMD does not

significantly affect the PMD limitation when chromatic dispersion is within this design guideline. This is evident from the fact that the limiting PMD values for the waveplate model and the first-order approximation for PMD agree within ten percent with $|\beta_2 L| \leq 1000 \text{ ps}^2$. From our results, we conclude that the presence of chromatic dispersion should not significantly reduce the PMD limit (relative to the limit with $\beta_2 L = 0$) if the chromatic dispersion is within normal design guidelines.

With $\beta_2 L = -2000 \text{ ps}^2$, the PMD limit of $0.126T$ is slightly lower than that with $\beta_2 L = 0$ and there is a large difference in the limiting PMD values for the waveplate model and first-order PMD. So for large values of chromatic dispersion higher order PMD has a significant effect on system performance and the PMD limit can be reduced relative to that with $\beta_2 L = 0$.

IV. CONCLUSION

Pulse distortion was examined for PMD in the presence of chromatic dispersion. The effective pulse broadening due to PMD was worsened due to the interaction of chromatic dispersion and higher order PMD. However, with the chromatic dispersion within normal design guidelines the PMD limit was not reduced by the presence of chromatic dispersion and, was within 10% of the limit given by the first-order approximation for PMD.

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